THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 1: **Real Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The LCM of 30, 42 and 66 is: 1

(a) 1155 **(b)** 6930 **(c)** 2310 **(d)** 1260

2. The least number that is divisible by every number from 1 to 6 is: 1

(a) 60 **(b)** 360 **(c)** 120 **(d)** 30

3. The largest number dividing 67, 103 and 175 leaving remainder 4 in each case is: 1

(a) 18 **(b)** 9 **(c)** 1 **(d)** 3

4. The smallest positive integer by which 12 must be multiplied to obtain a perfect square is: 1

(a) 6 **(b)** 3 **(c)** 12 **(d)** 2

5. Which of these is a rational number? 1

(a) $2\sqrt{5}$ **(b)** $\dfrac{\sqrt{12}}{\sqrt{3}}$ **(c)** $\sqrt{2}+\sqrt{3}$ **(d)** $\sqrt{4}$

6. A number ends with exactly 2 zeros if its prime factorisation (other factor 3) is: 1

(a) $2^3 \times 5^2 \times 3$ **(b)** $3^2 \times 7$ **(c)** $2^3 \times 5^1 \times 3$ **(d)** $2^1 \times 5^2 \times 3$

7. The LCM of 36, 48 and 72 is: 1

(a) 5184 **(b)** 1728 **(c)** 72 **(d)** 144

8. The least number that is divisible by every number from 1 to 7 is: 1

(a) 840 **(b)** 2520 **(c)** 210 **(d)** 420

9. The largest number dividing 75, 123 and 195 leaving remainder 3 in each case is: 1

(a) 24 **(b)** 12 **(c)** 48 **(d)** 6

10. The smallest positive integer by which 18 must be multiplied to obtain a perfect square is: 1

(a) 6 **(b)** 2 **(c)** 3 **(d)** 18

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Two numbers have HCF 12 and are in the ratio $3 : 5$. Find the numbers and their LCM. 2

12. Find the smallest positive integer that leaves remainders 7 and 13 when divided by 12 and 18 respectively. 2

13. Prove that $3\sqrt{7}$ is irrational, given that $\sqrt{7}$ is irrational. 2

14. Two numbers have HCF 15 and are in the ratio $4 : 7$. Find the numbers and their LCM. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A, B and C take 18, 24 and 36 minutes per round of a circular track. They start together. After how many minutes do they meet at the start again, and how many rounds has each completed? 3

16. Find the greatest number that divides 1251 and 9377 leaving remainder 1 in each case. 3

17. Is $5 \times 6 \times 7 + 7$ a composite number? Justify. 3

OR

A, B and C take 20, 28 and 35 minutes per round of a circular track. They start together. After how many minutes do they meet at the start again, and how many rounds has each completed? 3

18. Find the greatest number that divides 1157 and 1397 leaving remainder 5 in each case. 3

OR

Is $6 \times 7 \times 8 + 8$ a composite number? Justify. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Three sets of English, Hindi and Mathematics books have 96, 240 and 336 books. They are stacked so that each stack has the same number of books of one subject and all stacks have the same height. Find the minimum number of stacks. 5

OR

Prove that $\frac{1}{\sqrt{2}}$ is irrational and hence that $\frac{3}{\sqrt{2}}$ is irrational. 5

20. Three sets of English, Hindi and Mathematics books have 144, 192 and 288 books. They are stacked so that each stack has the same number of books of one subject and all stacks have the same height. Find the minimum number of stacks. 5

OR

Prove that $\frac{1}{\sqrt{3}}$ is irrational and hence that $\frac{5}{\sqrt{3}}$ is irrational. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND